

Vector Addition — Resultant of Vectors

Problem Set with Detailed Solutions · Class 11 — JEE Foundation

Instructions: For each problem find the resultant — its magnitude, and its direction wherever the question asks for it. Use the parallelogram / triangle law of vector addition (no components or axes). Angles are measured between the vectors when drawn from a common point. Take $\sqrt{2} \approx 1.41$, $\sqrt{3} \approx 1.73$ where needed. (26 problems)

Problems

- Two forces of 3 N and 4 N act at a point, at right angles (90°) to each other. Find the magnitude of their resultant.
- Two vectors, each of magnitude 10 units, are inclined to each other at 60° . Find the magnitude of their resultant.
- Two perpendicular vectors have magnitudes 5 units and 12 units. Find the magnitude of the resultant and the angle it makes with the 5-unit vector.
- Two forces of equal magnitude act at a point. If their resultant has the same magnitude as each force, find the angle between the two forces.
- Two vectors of magnitudes 8 N and 6 N are inclined at 60° . Find the magnitude of their resultant.
- Two vectors of magnitudes 7 units and 24 units are inclined at 120° . Find the magnitude of their resultant.
- For two vectors **A** and **B**, the magnitude of the sum **A** + **B** equals the magnitude of the difference **A** – **B**. Find the angle between **A** and **B**.
- Two equal vectors, each of magnitude 5 units, are inclined at 90° . Find the magnitude of their resultant and the angle it makes with either vector.
- A person walks 4 km due north and then 3 km due east. Find the magnitude of the net displacement and its direction (as an angle east of north).
- Two forces of 5 N and 3 N act at a point and have a resultant of magnitude 7 N. Find the angle between the two forces.
- Two vectors of magnitudes 2 units and 3 units are inclined at 30° . Find the magnitude of their resultant.
- Two vectors **A** and **B** have magnitudes 6 units and 10 units. Their resultant is found to be perpendicular to **A**. Find the angle between **A** and **B**, and the magnitude of the resultant.
- Three forces acting at a point keep it in equilibrium. Two of them are 8 N and 6 N, perpendicular to each other. Find the magnitude and direction of the third force.
- Two equal forces, each of magnitude P , are inclined at an angle θ . If their resultant has magnitude $P\sqrt{3}$, find θ .
- Two vectors of magnitudes 4 N and 4 N are inclined at 120° . Find the magnitude of their resultant.

- 16.** Two forces of 12 N and 5 N act at a point and give a resultant of 13 N. Are the two forces perpendicular? Justify your answer. Also find their resultant if the same two forces act in exactly opposite directions.
- 17.** Two equal vectors, each of magnitude 3 units, are inclined at 60° . Find the magnitude of the resultant and the angle it makes with either vector.
- 18.** Three vectors of equal magnitude are arranged so that each makes an angle of 120° with the next (they are placed symmetrically around a point). Find the magnitude of their resultant.
- 19.** Two perpendicular forces have magnitudes 10 N and $10\sqrt{3}$ N. Find the magnitude of the resultant and the angle it makes with the 10 N force.
- 20.** Two forces act at a point with 60° between them. One force is 6 N and their resultant is 14 N. Find the magnitude of the other force.
- 21.** A particle undergoes two successive displacements of 6 m and 10 m, with 60° between the two displacement vectors. Find the magnitude of the net displacement.
- 22.** Two vectors of magnitudes 3 units and 5 units give a resultant of 7 units. Find the angle between them. If this angle is then doubled, find the new magnitude of the resultant.
- 23.** Two forces of 5 N and 8 N are inclined at 60° . Find the magnitude of the resultant and the angle it makes with the 5 N force.
- 24.** Two forces, each of magnitude F , act at a point with angle θ between them. Find θ for which the resultant has magnitude (a) F , (b) $F\sqrt{2}$, (c) $F\sqrt{3}$, (d) $2F$.
- 25.** Two perpendicular vectors have magnitudes 9 units and 12 units. Find the magnitude of the resultant and the angle it makes with the 12-unit vector.
- 26.** Two equal forces, each of magnitude F , act at a point at 90° to each other. A third force, also of magnitude F , acts along the bisector of the angle between them (i.e. along their resultant). Find the magnitude of the resultant of all three forces.

Detailed Solutions

Key relations used throughout

- Magnitude of resultant of **A**, **B** with angle θ between them: $R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$
- Direction (angle α of the resultant from **A**): $\tan \alpha = (B \sin \theta) / (A + B \cos \theta)$
- Equal magnitudes ($A = B$): $R = 2A \cos(\theta/2)$, and the resultant bisects the angle
- Special cases: $\theta = 0^\circ \rightarrow R = A + B$ (maximum) ; $\theta = 180^\circ \rightarrow R = |A - B|$ (minimum) ; $\theta = 90^\circ \rightarrow R = \sqrt{A^2 + B^2}$

1. $R = \sqrt{3^2 + 4^2 + 2 \cdot 3 \cdot 4 \cdot \cos 90^\circ} = \sqrt{9 + 16 + 0} = \sqrt{25} = \mathbf{5 \text{ N}}$.
2. $R = \sqrt{10^2 + 10^2 + 2 \cdot 10 \cdot 10 \cdot \cos 60^\circ} = \sqrt{200 + 100} = \sqrt{300} = 10\sqrt{3} \approx \mathbf{17.32 \text{ units}}$. (Or $R = 2A \cos(\theta/2) = 2 \cdot 10 \cdot \cos 30^\circ = 10\sqrt{3}$.)
3. $R = \sqrt{5^2 + 12^2} = \sqrt{169} = \mathbf{13 \text{ units}}$. Direction from the 5-unit vector: $\tan \alpha = (12 \sin 90^\circ) / (5 + 12 \cos 90^\circ) = 12/5 = 2.4$, so $\alpha = \tan^{-1}(2.4) \approx \mathbf{67.4^\circ}$.
4. Equal magnitudes $\Rightarrow R = 2F \cos(\theta/2)$. Set $2F \cos(\theta/2) = F \Rightarrow \cos(\theta/2) = 1/2 \Rightarrow \theta/2 = 60^\circ \Rightarrow \theta = \mathbf{120^\circ}$.

Watch out: The resultant of two equal vectors equals either one only when they are 120° apart — worth memorising.

5. $R = \sqrt{8^2 + 6^2 + 2 \cdot 8 \cdot 6 \cdot \cos 60^\circ} = \sqrt{64 + 36 + 48} = \sqrt{148} = 2\sqrt{37} \approx \mathbf{12.17 \text{ N}}$.
6. $\cos 120^\circ = -1/2$. $R = \sqrt{7^2 + 24^2 + 2 \cdot 7 \cdot 24 \cdot (-1/2)} = \sqrt{49 + 576 - 168} = \sqrt{457} \approx \mathbf{21.38 \text{ units}}$.
7. $|\mathbf{A+B}| = |\mathbf{A-B}| \Rightarrow A^2 + B^2 + 2AB \cos \theta = A^2 + B^2 - 2AB \cos \theta \Rightarrow 4AB \cos \theta = 0 \Rightarrow \cos \theta = 0 \Rightarrow \theta = \mathbf{90^\circ}$.

Watch out: Sum and difference are equal in length exactly when the vectors are perpendicular (the parallelogram becomes a rectangle, whose diagonals are equal).

8. $R = \sqrt{5^2 + 5^2} = \sqrt{50} = 5\sqrt{2} \approx \mathbf{7.07 \text{ units}}$. The vectors are equal, so the resultant bisects the angle $\Rightarrow \mathbf{45^\circ}$ from each.
9. The two legs (north, then east) are perpendicular. Displacement $= \sqrt{4^2 + 3^2} = \sqrt{25} = \mathbf{5 \text{ km}}$. Direction: $\tan \beta = 3/4 \Rightarrow \beta = \tan^{-1}(0.75) \approx \mathbf{36.9^\circ \text{ east of north}}$.
10. $R^2 = A^2 + B^2 + 2AB \cos \theta \Rightarrow 49 = 25 + 9 + 2 \cdot 15 \cdot \cos \theta \Rightarrow 49 = 34 + 30 \cos \theta \Rightarrow \cos \theta = 15/30 = 1/2 \Rightarrow \theta = \mathbf{60^\circ}$.
11. $R = \sqrt{2^2 + 3^2 + 2 \cdot 2 \cdot 3 \cdot \cos 30^\circ} = \sqrt{13 + 12 \cdot (\sqrt{3}/2)} = \sqrt{13 + 6\sqrt{3}} \approx \sqrt{23.39} \approx \mathbf{4.84 \text{ units}}$.
12. "Resultant $\perp \mathbf{A}$ " means the component of the resultant along **A** is zero: $A + B \cos \theta = 0 \Rightarrow \cos \theta = -A/B = -6/10 = -0.6 \Rightarrow \theta = \mathbf{126.9^\circ}$. The resultant then equals the part perpendicular to **A**: $R = B \sin \theta = 10 \cdot 0.8 = \mathbf{8 \text{ units}}$. (Check: $\sqrt{36 + 100 - 72} = \sqrt{64} = 8$.)

Watch out: The correct condition is $A + B \cos \theta = 0$, not $B + A \cos \theta = 0$ — students often resolve along the wrong vector.

13. Resultant of the two perpendicular forces $= \sqrt{8^2 + 6^2} = 10 \text{ N}$, at $\tan^{-1}(6/8) \approx 36.9^\circ$ from the 8 N force. For equilibrium the third force must be equal and opposite: $\mathbf{10 \text{ N}}$, directed opposite to this resultant (at 180° to it).
14. Equal forces $\Rightarrow R = 2P \cos(\theta/2)$. Set $2P \cos(\theta/2) = P\sqrt{3} \Rightarrow \cos(\theta/2) = \sqrt{3}/2 \Rightarrow \theta/2 = 30^\circ \Rightarrow \theta = \mathbf{60^\circ}$.
15. Equal vectors at 120° : $R = 2 \cdot 4 \cdot \cos 60^\circ = 8 \cdot 1/2 = \mathbf{4 \text{ N}}$. (The resultant equals each force — exactly the 120° result again.)

16. $13^2 = 12^2 + 5^2 + 2 \cdot 12 \cdot 5 \cdot \cos \theta \Rightarrow 169 = 169 + 120 \cos \theta \Rightarrow \cos \theta = 0 \Rightarrow \theta = 90^\circ$, so **yes, perpendicular**. At $\theta = 180^\circ$: $R = |12 - 5| = 7 \text{ N}$.

Watch out: 12, 5, 13 is a right-triangle triple — that is the clue that the angle is 90° .

17. $R = 2 \cdot 3 \cdot \cos 30^\circ = 6 \cdot (\sqrt{3}/2) = 3\sqrt{3} \approx 5.20 \text{ units}$, bisecting the angle $\Rightarrow 30^\circ$ from each.
18. By symmetry the resultant is **0**. Reason: adding the first two (120° apart) gives a vector of magnitude $2a \cos 60^\circ = a$, pointing exactly opposite to the third vector, so it cancels it.
19. $R = \sqrt{(10^2 + (10\sqrt{3})^2)} = \sqrt{(100 + 300)} = \sqrt{400} = 20 \text{ N}$. Angle with the 10 N force: $\tan \alpha = (10\sqrt{3})/10 = \sqrt{3} \Rightarrow \alpha = 60^\circ$.
20. Let the other force be B. $14^2 = 6^2 + B^2 + 2 \cdot 6 \cdot B \cdot \cos 60^\circ \Rightarrow 196 = 36 + B^2 + 6B \Rightarrow B^2 + 6B - 160 = 0 \Rightarrow B = (-6 + \sqrt{676})/2 = (-6 + 26)/2 = 10 \text{ N}$. (Reject the negative root.)
21. $R = \sqrt{(6^2 + 10^2 + 2 \cdot 6 \cdot 10 \cdot \cos 60^\circ)} = \sqrt{(36 + 60)} = \sqrt{96} = 14 \text{ m}$.
22. $49 = 3^2 + 5^2 + 30 \cos \theta \Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \theta = 60^\circ$. Doubled: $\theta' = 120^\circ$, $\cos 120^\circ = -\frac{1}{2}$. $R' = \sqrt{(9 + 25 + 30 \cdot (-\frac{1}{2}))} = \sqrt{(34 - 15)} = \sqrt{19} \approx 4.36 \text{ units}$.

Watch out: Doubling the angle does NOT double or halve the resultant — it must be recomputed from the formula.

23. $R = \sqrt{(5^2 + 8^2 + 2 \cdot 5 \cdot 8 \cdot \cos 60^\circ)} = \sqrt{(25 + 40)} = \sqrt{65} \approx 8.06 \text{ N}$. Direction from the 5 N force: $\tan \alpha = (8 \sin 60^\circ)/(5 + 8 \cos 60^\circ) = 6.93/9 = 0.770 \Rightarrow \alpha \approx 37.6^\circ$.
24. Equal forces $\Rightarrow R = 2F \cos(\theta/2)$. (a) $F \Rightarrow \cos(\theta/2) = \frac{1}{2} \Rightarrow \theta = 120^\circ$. (b) $F\sqrt{2} \Rightarrow \cos(\theta/2) = 1/\sqrt{2} \Rightarrow \theta = 90^\circ$. (c) $F\sqrt{3} \Rightarrow \cos(\theta/2) = \sqrt{3}/2 \Rightarrow \theta = 60^\circ$. (d) $2F \Rightarrow \cos(\theta/2) = 1 \Rightarrow \theta = 0^\circ$ (vectors parallel).
25. $R = \sqrt{(9^2 + 12^2)} = \sqrt{225} = 15 \text{ units}$. Angle with the 12-unit vector: $\tan \alpha = 9/12 = 0.75 \Rightarrow \alpha = 36.9^\circ$.
26. The two equal forces at 90° give a resultant of $F\sqrt{2}$ along their bisector (45° from each). The third force F lies along this same bisector, so it simply adds on: total $= F\sqrt{2} + F = F(\sqrt{2} + 1) \approx 2.41 F$, along the bisector.

Watch out: Because the third force is collinear with the resultant of the first two, the magnitudes add directly — no cosine rule needed for the last step.